

Research on Medical Text Classification based on BioBERT-GRU-Attention

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Abstract—The growing sophistication of deep learning technology has driven advances in automated processing of medical texts. Applying deep learning technology to medical text classification can reduce the degree of manual intervention and improve the classification accuracy of medical texts. Therefore, this paper designs and proposes BGA and B2GA models based on BioBERT. The models use BioBERT as the word embedding layer, and use bidirectional GRU (BiGRU) and multi-head self-attention mechanism to extract feature vectors. Experimental verification: In the medical field, the classification effect of the BioBERT model is more accurate than the general BERT, and the accuracy rate is increased by about 38%; For datasets with fewer entries, BiGRU is more suitable than Bidirectional LSTM (BiLSTM); The BGA model proposed in this paper has an accuracy of more than 72% for automatic classification of medical texts, and the proposed B2GA model has an accuracy of more than 70% for automatic classification of medical texts. Compared with the BioBERT (proposed by Korea University) and Bio_ClinicalBERT model, they have a better text classification effect.

Keywords—Deep Learning, Medical Text Classification, BioBERT, BiGRU, BiLSTM, Multi-head Self-Attention Mechanism

I. INTRODUCTION

At present, with the improvement of medical technology and the development of science and technology, literature and other medical texts related to the medical field have shown an exponential growth trend in recent years [1], and the traditional classification of medical texts has also become more and more time-consuming, labor-intensive and low-quality shortcomings. At the same time, when classifying texts, one classification result is easily affected by the subjective consciousness of the classifying person, and the other is that the medical data to be classified also has the risk of information leakage. Therefore, it is extremely urgent and necessary to realize the automatic classification of medical texts.

In recent years, deep learning technology has developed very rapidly, and has achieved significant results in many

fields such as image recognition, speech recognition, machine translation and natural language processing (NLP). Applying deep learning to text classification realizes the transformation from the original manual feature extraction method to the automatic classification method based on deep learning. Therefore, it has become an inevitable trend to apply the NLP algorithm to the field of medical text classification by extracting medical text features through deep learning methods for automatic classification.

Text classification, as an important branch in the field of NLP, has always been one of the hotspots of scholars' research. The shallow classification models such as support vector machines, decision trees and random forests that appeared in the early stage require manual extraction of text features and are inefficient. Later, deep learning models such as RNN, Attention and BERT appeared, which made it possible to automatically extract text features and improved the efficiency and quality of text classification. Now, the models proposed by most scholars are fusion models that "enhance strengths and avoid weaknesses" for advanced models. For the extraction of biomedical relationships, in June 2020, Zhu Yu et al. proposed BioBERT and BiGRU models based on entity-aware attention mechanism to extract drug-drug interactions from biomedical literature [2]. In view of the problems of low utilization rate and difficulty in extracting classification information from Chinese medical records, Du Lin et al. proposed a short text classification model of medical records fused by BERT+Bi-LSTM+Attention in 2020 [3]. Zhao Yang et al. also conducted related research on Chinese medicine, and they proposed a Chinese medical literature classification based on the BERT model [4]. Aiming at the problem of Chinese electronic medical record text classification, Li Chaofan and others from Xuzhou Medical University proposed a combination of CNN-BiLSTM model based on attention mechanism, and the F1 value is as high as 97.85% [5].

This paper draws on the experience of predecessors to accurately classify medical texts by integrating models such as BioBERT [6], multi-head self-attention mechanism [7] and BiGRU [8].

II. INTRODUCE RELATED MODELS

According to the attention architecture LEAM [9] proposed by Guoyin Wang et al. in 2018, it can be seen that text classification needs to go through the classic three steps. First, the distributed expression matrix C of the text is converted into the word vector WE through the embedding layer. Then, WE is transformed into a fixed-length feature vector Vec by suitable means. Finally, the Vec is processed by the classifier into a vector representation of the same length as the label.

A. BioBERT

For the human language system, words are the basic unit of meaning in natural language, and the reason why humans can recognize and understand them is because in the long-term or imperceptible learning, the human brain establishes complex mapping connections. For computers, if you want to realize this kind of mapping connection, you need a reasonable operation process. The basis of this process is text encoding, or word embeddings, which represent text matrices as numerical vectors. The quality of word embedding determines whether the prediction result of the model is reliable, because word embedding is one of the important foundations of model training.

The earliest text encoding technology is One-hot encoding, which is characterized by extremely sparse encoding matrix and many shortcomings.

Then came the epoch-making Word2vec and GloVe. Among them, the Word2vec tool has two forms, one is a CBOW model that uses context to infer intermediate words, and the other is a Skip-gram model that infers context words from intermediate words. The characteristic of this type of method is that the vector after word embedding is static, and it cannot effectively express polysemy. In the medical field, foreign language abbreviations with polysemy are extremely common, such as: ICH in the medical field includes

intracerebral hematoma, intracerebral hemorrhage, intracranial hemotoma, intracranial hemorrhage and so on [10].

Then in mid-October 2018, the Google team proposed the Transformer-based BERT [11] pre-training model. When it came out, it was famous in the field of deep learning for refreshing 11 NLP tasks including SWAG and GLUE. The BERT model is a way of dynamically representing word embeddings, which can solve the problem that the above encoding tools cannot express polysemy. This model is a bidirectional Transformer pre-training model that integrates the idea of the CBOW model, the structure of the ELMo model and the Transformer component of the OpenAI GPT model.

In addition to using the case-sensitive general BERT officially released by Google, this paper also uses the Bio_ClinicalBERT pre-training model [12] proposed by the Emily Alsentzer team in 2019 and the dmiis-lab/biobert-base-cased-v1.2 (Dmiis_Biobert) pre-training model [13] proposed by the Jinhyuk Lee team in 2020. Validation and comparison experiments are carried out in Section 4.4.1.

B. Multi-Head Self-Attention Mechanism

In recent years, the popular models and architectures, such as BERT and Transformer, etc., the attention mechanism [7] is one of the core components. The attention mechanism is a method that maps a d_q -dimensional vector Query and a pair of d_k -dimensional and d_v -dimensional Key-Value pairs to obtain the result, which is calculated as formula (1). In order to enable the model of this paper to comprehensively and comprehensively consider the effective information of words at different positions, this paper adopts the multi-head self-attention mechanism, and its calculation method is as follows from formulas (2) to (3).

$$Attention(Query, Key, Value) = softmax\left(\frac{Query \odot Key}{\sqrt{d_k}}\right) \odot Value \quad (1)$$

$$MultiHead(Query, Key, Value) = Concat(head_1, \dots, head_i, \dots, head_h)W^O \quad (2)$$

$$head_i = Attention(Query \odot W_i^{Query}, Key \odot W_i^{Key}, Value \odot W_i^{Value}) \quad (3)$$

$W_i^{Query} \in R^{d_{model} \times d_k}$, $W_i^{Key} \in R^{d_{model} \times d_k}$, $W_i^{Value} \in R^{d_{model} \times d_v}$, $W^O \in R^{head \times d_v \times d_{model}}$ in the above formula.

represents the hidden state of the forward GRU at time i , and $\overleftarrow{h}_i$ represents the hidden state of the reverse GRU at time i .

$$\overrightarrow{h}_i = GRU(\overrightarrow{h}_{i-1}, x_i) \quad (4)$$

$$\overleftarrow{h}_i = GRU(\overleftarrow{h}_{i-1}, x_i) \quad (5)$$

$$h_i = W \begin{bmatrix} \overrightarrow{h}_i \\ \overleftarrow{h}_i \end{bmatrix} + b_i \quad (6)$$

W in formula (6) represents the weight matrix of $\overrightarrow{h}_i$ and $\overleftarrow{h}_i$ corresponding to the BiGRU at time i , and b_i represents the bias of the hidden state h_i at the current moment.

III. BGA AND B2GA MODELS

This paper proposes the BGA model and the B2GA model. The logical structure is shown in Fig. 1. The neural network

C. Bidirectional Gated Recurrent Unit

Gated Recurrent Unit (GRU) [8] is one of many variants in RNN [14], which is similar to LSTM but without output gates. GRUs use fewer parameters than LSTMs and are more efficient than LSTMs when dealing with datasets with fewer entries [15, 16]. It can solve the gradient vanishing problem of traditional RNN by update gate and reset gate, and GRU can selectively retain upstream information and delete interference information that is irrelevant to model prediction.

This paper uses BiGRU (Bidirectional GRU), which is the result of the forward GRU and reverse GRU values processed by a connection layer, which can learn context information from both sides of the input medical text. The calculation of the BiGRU model is shown in formulas (4) to (6). $\overrightarrow{h}_i$

layer is composed of BiGRU and attention mechanism. The detailed structure is shown in Fig. 2 and Fig. 3. The two models are mainly slightly different from the neural network layer, which is mainly reflected in the input sequence of the multi-head self-attention mechanism layer. The BGA model adds the BiGRU layer and the multi-head self-attention

mechanism layer after the word embedding layer based on BioBERT, which is a series of the fusion network model. In the B2GA model, the input sequence of the multi-head self-attention mechanism layer is the result of splicing the hidden layer states of two parallel BiGRUs.

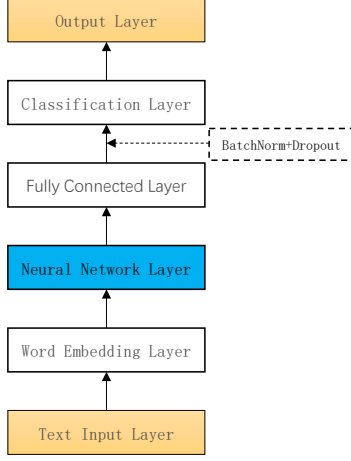


Fig. 1. Model logic structure

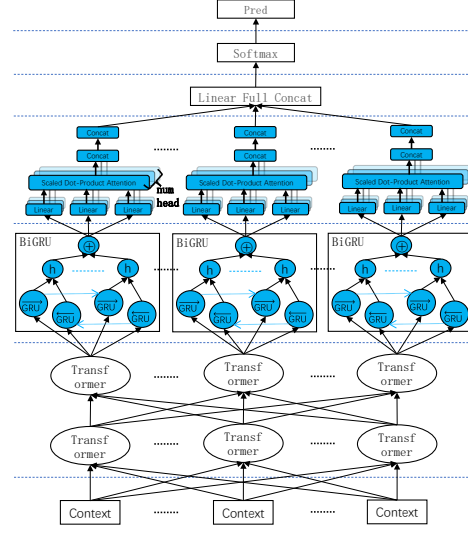


Fig. 2 BGA model structure

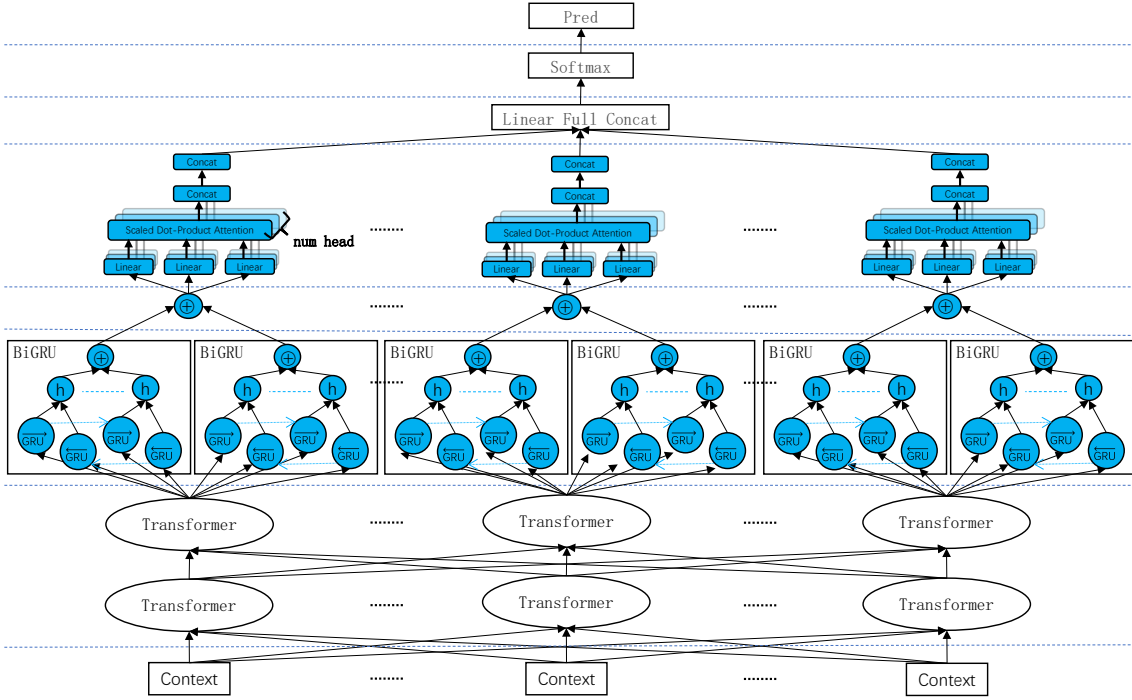


Fig. 3. B2GA model structure

IV. EXPERIMENTAL REALIZATION

A. Experimental Environment

The environment configuration of the experiment in this

paper is shown in Table I. The Python scientific computing framework used is Pytorch, which is more flexible, because it can not only replace NumPy seamlessly, but also accelerate the calculation of tensor through GPU.

TABLE I. EXPERIMENTAL ENVIRONMENT CONFIGURATION

	Environment	Processor	RAM	Development Tools	Graphics Driver	Development Platform	Experimental Framework
Name And Version	Windows 10	Intel(R) Core(TM) i5-8250U CPU @1.60GHz 1.80GHz	8.00GB	Python 3.9.7	NVIDIA CUDA 11.5	PyCharm 2021.2.3	Pytorch 1.11

B. Experimental Data Set and Experiment Evaluation Metrics

1) Experimental Dataset

Currently publicly available medical data is extremely difficult to find due to the HIPAA privacy regulations. The raw data for the experiments in this paper were obtained from mtsamples.com and compiled by analytical engineer Tara Boyle, a dataset detailing medical transcription samples from a variety of specialties.

The data set mtsamples compiled by engineer Tara Boyle contains a total of 4998 pieces of data, which are divided into 40 labels with different entries such as Allergy / Immunology, Autopsy and Bariatrics.

After a series of professional processing on the mtsamples data set, there are still 7 types of data such as Cardiovascular / Pulmonary (C/P), Gastroenterology (G), Neurology (N), Obstetrics / Gynecology (O/G), Orthopedic (O), Radiology (R) and Urology (U). Finally, the processed data set is divided into 8:1:1 training set, validation set and test set. The class distribution after division is shown in Fig. 4.

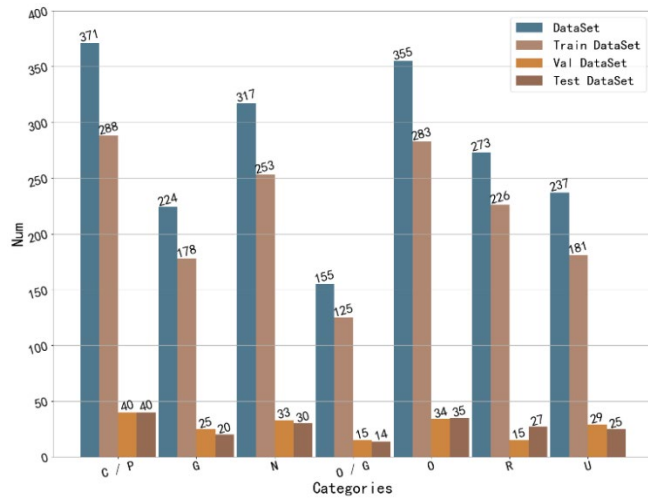


Fig. 4. The result of dividing the mtsamples dataset

2) Evaluation Index of the Experiment

According to the prediction results of the experiment, the evaluation index analysis table as shown in Table II can be generated, which indicates that TP and TN are correctly predicted, and FP and FN are wrongly predicted.

TABLE II. EVALUATION INDEX ANALYSIS

Evaluation Indicators	Actual Positive	Actually Negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

For the model, this paper mainly uses the Accuracy, Precision, Recall and F1 Score as evaluation indicators to evaluate the experimental results. The main purpose is to adjust the hyperparameters when training the model, it is also used to evaluate the performance of the whole model. For each class of the data set, this paper uses the Precision (P), Recall (R) and F1 Score (F1) as the criteria for evaluating the quality of the model. The calculation methods of the above evaluation indicators are as formulas (7) to (10):

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

$$F1-Score = \frac{2Precision \cdot Recall}{Precision+Recall} \quad (10)$$

TABLE III. DROPOUT AND LEARNING RATE CONFIGURATION TABLE

Model	Dropout	Learning Rate
bert-base-cased	0.08	1e-3
Bio_ClinicalBERT	0.5	1e-2
Dmis_Biobert	0.5	5e-3
BA	0.3	8e-4
BG	0.5	1e-3
BLA	0.4	8e-3
BAG	0.5	1e-3
BGA	0.4	8e-3
BA2G	0.5	5e-4
B2GA	0.4	8e-3

C. Hyperparameters for Experiment Running

In order to reduce the influence of unnecessary factors as much as possible in the model comparison experiment, some hyperparameters of each model are uniformly configured in the experiment. The specific configuration information is shown in Table III. Among them, only the number of epochs in experiment 1 is 50, and the other experiments are 30.

TABLE IV. UNIFIED HYPERPARAMETER CONFIGURATION TABLE FOR EACH MODEL

Model Layer Name	Hyperparameters	Values
Biobert Word Embedding Layer	bert-base-cased Embedding Size	768
	Bio_ClinicalBERT Embedding Size	768
	Dmis_Biobert Embedding Size	768
	Maximum Sentence Length	128
Attention Layer	Parallel Attention Heads	8
BiGRU/BiLSTM Layer	Hidden Layer Size	32
	Model Layers	16
Classifier Layer	Output Size	7
Train The Model	Batch Size	64
	Epochs	30/50

The last is the configuration combination of dropout and learning rate for each model. The specific parameters are shown in Table IV. Most of the combinations are obtained through experimental screening.

D. Experimental Results and Analysis

This paper sets up three sets of experiments to evaluate the accuracy and efficiency of BGA and B2GA models by comparing with different model structures.

Experiment 1 aims to verify that BioBERT as a word embedding layer of the model has advantages over general BERT. Experiment 2 aims to test the accuracy and stability of the BGA and B2GA models compared with other benchmark models. The last experiment aims to verify the importance and necessity of related components in the structure of BGA and B2GA models.

1) BERT Word Embedding Layer Experiment

BioBERT is a medical domain-specific language representation model pretrained on a large-scale medical corpus [2]. In this paper, three BERT pre-training models are selected for testing experiments.

For the experimental results of the three BERTs on the mtsamples validation set, see Fig. 5 for the first 14 epochs. Observing the experimental results of each model on the validation set, it is found that the performance of the general

BERT is not satisfactory compared to the performance of the two BioBERTs. The Loss curve declines slowly and the performance is not excellent enough. The evaluation indicators such as accuracy rate are not as good as BioBERT.

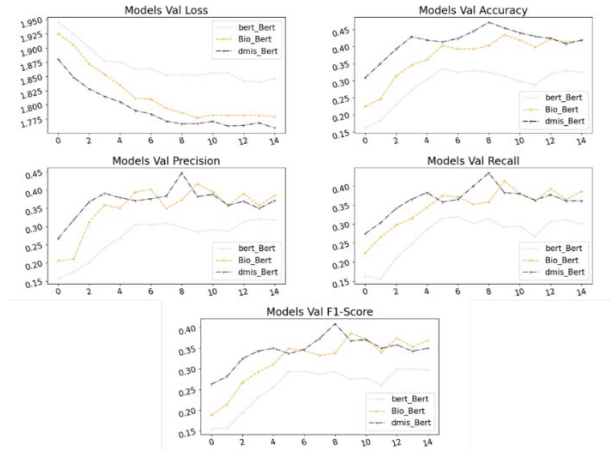


Fig. 5. Experimental Results of The BERT Model on The Validation Set

Now analyze the experimental results of the three BERT models in Table V on the test set. The Best Val Loss column represents the optimal Loss value of the model on the

validation set, and Epoch is the epochs value corresponding to the optimal Loss value.

TABLE V. THE TRAINING RESULTS OF THE PRE-TRAINED MODEL ON THE MTSAMPLES DATASET

BERT pre-training	Accuracy	Precision	Recall	F1 Score	Best Val Loss	Epoch	Time(SECS)
Bert-base-cased	0.31	0.28	0.30	0.28	1.83	33	42
Bio_ClinicalBERT	0.43	0.42	0.40	0.40	1.76	31	48
Dmis_Biobert	0.42	0.45	0.42	0.43	1.75	34	31

Looking at Table V, it can be seen that Dmis_Biobert improves the accuracy by 35% compared with general BERT, and Bio_ClinicalBERT increases by 39%, which also confirms that there are obstacles to the application of general BERT in the medical field.

Through the above analysis and comparison, it is found that BioBERT can significantly improve the accuracy of the medical text classification model. When used alone, the Bio_ClinicalBERT model proposed by Emily's team performs slightly better than the Dmis_Biobert model proposed by

Jinhyuk's team on the medical text dataset. Therefore, in the follow-up experiments, this paper will only use two kinds of BioBERT as the word embedding layer of the model to conduct experiments.

2) Experiments of Each Model Based On BioBERT

In addition to the BioBERT models of Bio_ClinicalBERT and Dmis_Biobert, this paper also uses BAG (BioBERT + Attention + BiGRU) and BA2G (BioBERT + Attention + two-layer BiGRU) as the benchmark models, and the corresponding prediction results are shown in Table VI.

TABLE VI. PREDICTION RESULTS FOR THE MTSAMPLES DATASET

BioBERT Model Types (Word Embedding Layer)	Evaluation Indicators	BAG	BGA	BA2G	B2GA
-	number of model parameters	2798870	440726	2798902	453174
Bio_ClinicalBERT	Accuracy	0.701	0.696	0.696	0.701
	F1 Score	0.674	0.679	0.680	0.707
	Recall	0.682	0.697	0.702	0.710
	Precision	0.713	0.709	0.694	0.718
Dmis_Biobert	Accuracy	0.712	0.723	0.717	0.717
	F1 Score	0.675	0.705	0.691	0.696
	Recall	0.705	0.721	0.712	0.717
	Precision	0.708	0.715	0.692	0.709

By comparison, it is found that the number of parameters of the BAG and BA2G models is about 6.2 times that of the BGA and B2GA models, because the medical text data set used in this paper is relatively small, if it is applied to a larger medical text data set, when the model runs, The huge number of parameters will consume a lot of system resources and will seriously reduce the system performance. Therefore, under the

same experimental environment conditions, the BGA and B2GA models are better than the BAG and BA2G models. In terms of the experimental Accuracy, Recall and Precision evaluation indicators, the BGA and B2GA models perform slightly better than the BAG and BA2G models, and the F1 values of the BGA and B2GA models are relatively high, indicating that the BGA and B2GA models are not only in

terms of performance, but also It is also excellent in stability and accuracy.

Looking at the BioBERT models of the two word embedding layers as a whole, the fusion model based on Dmis_Biobert has higher accuracy, higher F1 value and better stability than the fusion model based on Bio_ClinicalBERT. This shows that Dmis_Biobert is more suitable than Bio_ClinicalBERT as a word embedding layer for a small medical text classification model like this article. The reason may be that Bio_ClinicalBERT is mainly aimed at the

electronic health records of hospital ICU patients. For a comprehensive medical text dataset such as mtsamples, it may be Not friendly.

Table VII shows the prediction results of the four models for each category. The classification of the R-type data is generally poor. The best is the BAG model based on Bio_ClinicalBERT, with an accuracy rate of 64%, but the F1 value is about 33%. It means that the model's prediction for this category is not stable.

TABLE VII. MODEL PREDICTION RESULTS FOR EACH CATEGORY

Model	Evaluation Indicators	Bio ClinicalBERT							Dmis Biobert						
		C/P	G	N	O/G	O	R	U	C/P	G	N	O/G	O	R	U
BAG	P	0.74	0.88	0.55	0.79	0.64	0.67	0.76	0.77	0.89	0.68	0.79	0.67	0.4	0.78
	R	0.93	0.7	0.57	0.79	0.77	0.22	0.88	0.93	0.8	0.5	0.79	0.8	0.22	1
	F1	0.82	0.78	0.56	0.79	0.7	0.33	0.81	0.84	0.84	0.58	0.79	0.73	0.29	0.88
BGA	P	0.82	0.88	0.53	0.76	0.62	0.5	0.75	0.82	0.85	0.64	0.8	0.65	0.41	0.81
	R	0.82	0.75	0.57	0.93	0.8	0.22	0.84	0.93	0.85	0.53	0.86	0.8	0.26	0.84
	F1	0.82	0.81	0.55	0.84	0.7	0.31	0.79	0.87	0.85	0.58	0.83	0.72	0.32	0.82
BA2G	P	0.91	0.76	0.52	0.73	0.68	0.5	0.74	0.81	0.89	0.6	0.75	0.66	0.47	0.79
	R	0.75	0.8	0.53	0.79	0.74	0.41	0.92	0.88	0.8	0.6	0.86	0.77	0.26	0.88
	F1	0.82	0.78	0.52	0.76	0.71	0.45	0.82	0.84	0.84	0.6	0.8	0.71	0.33	0.83
B2GA	P	0.83	0.94	0.54	0.76	0.65	0.37	0.87	0.8	0.77	0.6	0.83	0.68	0.5	0.79
	R	0.88	0.85	0.43	0.93	0.74	0.37	0.8	0.88	0.85	0.7	0.71	0.74	0.22	0.88
	F1	0.85	0.89	0.48	0.84	0.69	0.37	0.83	0.83	0.81	0.65	0.77	0.71	0.31	0.83

In terms of overall performance, BGA and B2GA are on a par with each other. However, experiments have shown that the combination of BGA and Dmis_Biobert is better, and the fusion of B2GA and Bio_ClinicalBERT is more suitable. B2GA increases the accuracy from 42% to over 72% relative to Bio_ClinicalBERT, and B2GA increases the accuracy from 43% to over 70% relative to Dmis_Biobert.

3) Ablation Studies

This section will study the role of BiGRU and multi-head self-attention mechanism in BGA and B2GA models. The comparison models are BA (BioBERT + Attention), BG (BioBERT + BiGRU) and BLA (BioBERT + BiLSTM + Attention). Detailed results of the experiment as shown in Table VIII and Table IX.

TABLE VIII. PREDICTION RESULTS OF BA, BG AND BLA MODELS ON MEDICAL TEXT DATASET MTSAMPLES

BioBERT Model Types (Word Embedding Layer)	Evaluation Indicators	BA	BG	BLA
-	number of model parameters	2364054	436502	586134
Bio_ClinicalBERT	Accuracy	0.60	0.69	0.63
	F1 Score	0.60	0.70	0.54
	Recall	0.62	0.71	0.57
	Precision	0.60	0.70	0.55
Dmis_Biobert	Accuracy	0.63	0.67	0.66
	F1 Score	0.63	0.66	0.59
	Recall	0.66	0.69	0.62
	Precision	0.61	0.66	0.61

TABLE IX. PREDICTION RESULTS OF BA, BG AND BLA MODELS FOR EACH CATEGORY

Model	Evaluation Indicators	Bio ClinicalBERT							Dmis Biobert						
		C/P	G	N	O/G	O	R	U	C/P	G	N	O/G	O	R	U
BA	P	0.76	0.68	0.46	0.75	0.57	0.13	0.81	0.78	0.81	0.47	0.76	0.63	0.1	0.81
	R	0.78	0.75	0.43	0.86	0.57	0.11	0.84	0.8	0.85	0.47	0.93	0.63	0.07	0.84
	F1	0.77	0.71	0.45	0.8	0.57	0.12	0.82	0.79	0.83	0.47	0.84	0.63	0.08	0.82
BG	P	0.91	0.77	0.48	0.8	0.63	0.42	0.88	0.76	0.83	0.55	0.72	0.68	0.31	0.86
	R	0.78	0.85	0.47	0.86	0.74	0.41	0.84	0.78	0.75	0.53	0.93	0.74	0.3	0.72
	F1	0.84	0.81	0.47	0.83	0.68	0.42	0.86	0.77	0.79	0.54	0.81	0.71	0.3	0.78
BLA	P	0.69	0.86	0.54	0	0.59	0.55	0.64	0.77	0.88	0.53	0.75	0.59	0	0.65
	R	0.88	0.6	0.43	0	0.83	0.41	0.84	0.93	0.7	0.6	0.43	0.86	0	0.88
	F1	0.77	0.71	0.48	0	0.69	0.47	0.72	0.84	0.78	0.56	0.55	0.7	0	0.75

According to the experimental results, except for the BG model, the other two models have more model parameters than the BGA and B2GA models, and the performance and stability of the BA, BG, and BLA models are also lower than those of the BGA and B2GA models. Experiments show that the effect of using BiGRU for feature extraction first is more efficient and more stable than the effect of using multi-head self-attention mechanism to obtain attention weight first, and using BiGRU for feature extraction of medical text is more efficient than using BiLSTM to extract features.

V. CONCLUSION AND OUTLOOK

In this paper, a deep learning method is used to construct a neural network model based on BioBERT, multi-head self-attention mechanism and BiGRU to classify medical texts. Through the above-mentioned comparative experiments, it can be observed that the model architecture proposed is effective and excellent, and can effectively improve the accuracy of medical text classification. It is reliable to use BioBERT as the word embedding layer to obtain the input sequence of the model. It is necessary and effective to extract

features through the multi-head self-attention mechanism and double-layer bidirectional GRU, and finally the results obtained by the classifier are accurate. Therefore, the BGA and B2GA models proposed in this paper are reasonable and efficient.

Although the experiments in this paper have reached a considerable level, there are still further studies on these model, mainly in the following points:

- Although the overall accuracy of the two models proposed in this paper has reached the expected standard, the classification accuracy of individual categories, such as radiology category R, is low, which is the key improvement part in the next step. There are three reasons for the current analysis: First, it may be that the R category processed in this paper is special, and its number in the validation set is about half of that in the test set; second, the two types of BioBERT used in this paper may not meet the expected requirements for the encoding effect of class R; third, there may be problems with the R-type data in the medical text dataset mtsamples, which may not be suitable for classification prediction, or the classification labels are set unreasonably.
- The dataset used in this experiment is small, and the next step is to obtain a larger dataset and conduct experiments on it.
- For some reasons, the English medical text dataset is used in the experiment in this paper, and the next step is to conduct classification research on Chinese medical texts.

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